

Finite Difference Method for Hyperbolic Equations

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1 Hyperbolic Partial Differential Equations

The hyperbolic partial differential equation, also known as the wave equation, is given by

$$\frac{\partial^2 u}{\partial t^2}(x, t) - \alpha^2 \frac{\partial^2 u}{\partial x^2}(x, t) = 0, \quad 0 < x < l, \quad t > 0. \quad (1)$$

It is subject to the boundary conditions

$$u(0, t) = u(l, t) = 0, \quad \text{for } t > 0, \quad (2)$$

and the initial conditions

$$u(x, 0) = f(x), \quad \frac{\partial u}{\partial t}(x, 0) = g(x), \quad \text{for } 0 \leq x \leq l, \quad (3)$$

where α is a constant determined by the physical conditions of the problem. The wave equation is ubiquitous throughout physics.

1.1 Finite Difference Method for the Wave Equation

As in the finite-difference methods used to solve parabolic and elliptic equations, an integer $m > 0$ is selected to define lattice points along the x -axis, with $h = l/m$. A temporal step size $k > 0$ is also selected. The grid points (x_i, t_j) are defined by

$$x_i = ih, \quad t_j = jk, \quad (4)$$

for $i = 0, 1, 2, \dots, m$ and $j = 0, 1, 2, \dots$

At every interior grid point, namely, every point that does not lie on the grid boundary, the wave equation in Eq. ?? becomes

$$\frac{\partial^2 u}{\partial t^2}(x_i, t_j) - \alpha^2 \frac{\partial^2 u}{\partial x^2}(x_i, t_j) = 0. \quad (5)$$

The finite-difference method is obtained by approximating the second partial derivatives with centered-difference quotients; see Ref. [?]. These approximations are

$$\frac{\partial^2 u}{\partial t^2}(x_i, t_j) = \frac{u(x_i, t_{j+1}) - 2u(x_i, t_j) + u(x_i, t_{j-1}))}{k^2} - \frac{k^2}{12} \frac{\partial^4 u}{\partial t^4}(x_i, t_j), \quad (6)$$

where $\mu_j \in (t_{j-1}, t_{j+1})$, and

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_j) = \frac{u(x_{i+1}, t_j) - 2u(x_i, t_j) + u(x_{i-1}, t_j))}{h^2} - \frac{h^2}{12} \frac{\partial^4 u}{\partial x^4}(\xi_i, t_j), \quad (7)$$

where $\xi_i \in (x_{i-1}, x_{i+1})$. Substituting Eqs. ?? and ?? into Eq. ?? gives

$$\begin{aligned} & \frac{u(x_i, t_{j+1}) - 2u(x_i, t_j) + u(x_i, t_{j-1}))}{k^2} - \alpha^2 \frac{u(x_{i+1}, t_j) - 2u(x_i, t_j) + u(x_{i-1}, t_j))}{h^2} \\ &= \frac{1}{12} \left[k^2 \frac{\partial^4 u}{\partial t^4}(x_i, \mu_j) - \alpha^2 h^2 \frac{\partial^4 u}{\partial x^4}(\xi_i, t_j) \right]. \end{aligned} \quad (8)$$

Neglecting the error term involving the fourth derivatives in time and space yields the difference equation

$$\frac{w_{i,j+1} - 2w_{i,j} + w_{i,j-1}}{k^2} - \alpha^2 \frac{w_{i+1,j} - 2w_{i,j} + w_{i-1,j}}{h^2} = 0. \quad (9)$$

Defining

$$\lambda = \frac{\alpha k}{h}, \quad (10)$$

Eq. ?? can be rewritten as

$$w_{i,j+1} - 2w_{i,j} + w_{i,j-1} - \lambda^2 w_{i+1,j} + 2\lambda^2 w_{i,j} - \lambda^2 w_{i-1,j} = 0. \quad (11)$$

Solving for the approximation at the next time level gives

$$w_{i,j+1} = 2(1 - \lambda^2)w_{i,j} + \lambda^2 (w_{i+1,j} + w_{i-1,j}) - w_{i,j-1}. \quad (12)$$

This equation is valid for $i = 1, 2, \dots, m-1$ and $j = 1, 2, \dots$. The boundary conditions give

$$w_{0,j} = w_{m,j} = 0, \quad \forall j = 1, 2, 3, \dots, \quad (13)$$

and the initial displacement implies

$$w_{i,0} = f(x_i), \quad \forall i = 1, 2, \dots, m-1. \quad (14)$$

For the standard explicit scheme, stability requires the Courant–Friedrichs–Lewy condition $\lambda \leq 1$.

The recurrence can be written conveniently in matrix form:

$$\begin{bmatrix} w_{1,j+1} \\ w_{2,j+1} \\ w_{3,j+1} \\ \vdots \\ w_{m-1,j+1} \end{bmatrix} = \begin{bmatrix} 2(1-\lambda^2) & \lambda^2 & 0 & \dots & 0 \\ \lambda^2 & \ddots & \ddots & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \lambda^2 \\ 0 & \dots & 0 & \lambda^2 & 2(1-\lambda^2) \end{bmatrix} \begin{bmatrix} w_{1,j} \\ w_{2,j} \\ w_{3,j} \\ \vdots \\ w_{m-1,j} \end{bmatrix} - \begin{bmatrix} w_{1,j-1} \\ w_{2,j-1} \\ w_{3,j-1} \\ \vdots \\ w_{m-1,j-1} \end{bmatrix}. \quad (15)$$

Equations ?? and ?? show that the $(j+1)$ th time step requires knowledge of the solutions at time levels j and $j-1$. This creates an initialization problem: the values at $j=0$ are given by the initial displacement, but the values at $j=1$, which are needed to compute $w_{i,2}$, must be obtained from the initial velocity condition

$$\frac{\partial u}{\partial t}(x, 0) = g(x), \quad 0 \leq x \leq l. \quad (16)$$

One approach is to replace $\partial u / \partial t$ with a forward-difference approximation:

$$\frac{\partial u}{\partial t}(x_i, 0) = \frac{u(x_i, t_1) - u(x_i, 0)}{k} - \frac{k}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \tilde{\mu}_i), \quad (17)$$

for some $\tilde{\mu}_i \in (0, t_1)$. Solving for $u(x_i, t_1)$ gives

$$\begin{aligned} u(x_i, t_1) &= u(x_i, 0) + k \frac{\partial u}{\partial t}(x_i, 0) + \frac{k^2}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \tilde{\mu}_i) \\ &= u(x_i, 0) + kg(x_i) + \frac{k^2}{2} \frac{\partial^2 u}{\partial t^2}(x_i, \tilde{\mu}_i). \end{aligned} \quad (18)$$

Omitting the truncation error gives the approximation

$$w_{i,1} = w_{i,0} + kg(x_i), \quad \forall i = 1, 2, \dots, m-1. \quad (19)$$

However, this approximation has truncation error $O(k)$, whereas the centered recurrence in Eq. ?? is second-order accurate in time.

1.2 Improving the Initial Approximation

To obtain a more accurate approximation for $u(x_i, t_1)$, expand it in a second-order Maclaurin polynomial in t :

$$u(x_i, t_1) = u(x_i, 0) + k \frac{\partial u}{\partial t}(x_i, 0) + \frac{k^2}{2} \frac{\partial^2 u}{\partial t^2}(x_i, 0) + \frac{k^3}{6} \frac{\partial^3 u}{\partial t^3}(x_i, \hat{\mu}_i), \quad (20)$$

for some $\hat{\mu}_i \in (0, t_1)$. If f'' exists, then

$$\frac{\partial^2 u}{\partial t^2}(x_i, 0) = \alpha^2 \frac{\partial^2 u}{\partial x^2}(x_i, 0) = \alpha^2 \frac{\partial^2 f}{\partial x^2}(x_i) = \alpha^2 f''(x_i). \quad (21)$$

Therefore,

$$u(x_i, t_1) = u(x_i, 0) + kg(x_i) + \frac{\alpha^2 k^2}{2} f''(x_i) + \frac{k^3}{6} \frac{\partial^3 u}{\partial t^3}(x_i, \hat{\mu}_i). \quad (22)$$

This gives the approximation

$$w_{i,1} = w_{i,0} + kg(x_i) + \frac{\alpha^2 k^2}{2} f''(x_i), \quad (23)$$

with a local truncation error of order $O(k^3)$.

If $f \in C^4[0, l]$ but $f''(x_i)$ is not directly available, the centered-difference formula

$$f''(x_0) = \frac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2} - \frac{h^2}{12} f^{(4)}(\xi) \quad (24)$$

can be used for some ξ satisfying $x_0 - h < \xi < x_0 + h$. Thus,

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2} - \frac{h^2}{12} f^{(4)}(\tilde{\xi}_i), \quad (25)$$

for some $\tilde{\xi}_i \in (x_{i-1}, x_{i+1})$. It follows that

$$u(x_i, t_1) = u(x_i, 0) + kg(x_i) + \frac{k^2 \alpha^2}{2h^2} [f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))] + O(k^3 + h^2 k^2). \quad (26)$$

Using $\lambda = \alpha k/h$ gives

$$\begin{aligned} u(x_i, t_1) &= u(x_i, 0) + kg(x_i) + \frac{\lambda^2}{2} [f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))] + O(k^3 + h^2k^2) \\ &= (1 - \lambda^2)f(x_i) + \frac{\lambda^2}{2}f(x_{i+1}) + \frac{\lambda^2}{2}f(x_{i-1}) + kg(x_i) + O(k^3 + h^2k^2). \end{aligned} \quad (27)$$

The improved initialization formula is therefore

$$w_{i,1} = (1 - \lambda^2)f(x_i) + \frac{\lambda^2}{2}f(x_{i+1}) + \frac{\lambda^2}{2}f(x_{i-1}) + kg(x_i). \quad (28)$$

It can be used to compute $w_{i,1}$ for every $i = 1, 2, \dots, m-1$. The subsequent approximations are obtained from the system in Eq. ??.

2 Finite Difference Algorithm for the Wave Equation

The development in the preceding section provides the tools needed to construct an algorithm for hyperbolic partial differential equations of the form in Eq. ?. The procedure is as follows:

- Construct the matrix system in Eq. ?.
- Use the improved initial approximation in Eq. ? to compute $w_{i,1}$.
- Apply the matrix recurrence for the required number of time steps.

2.1 Algorithm

The following pseudocode implements the explicit centered finite-difference scheme together with the improved first time step.

Algorithm 1 Explicit finite-difference method for the wave equation

```
1: function FINITE_DIFFERENCE_METHOD_HYPERBOLIC( $l, T, \alpha, N, m, f, g$ )
2:    $h \leftarrow l/m$ 
3:    $k \leftarrow T/N$ 
4:    $\lambda \leftarrow \alpha k/h$ 
5:    $x \leftarrow \text{linspace}(0, l, m+1)$ 
6:    $t \leftarrow \text{linspace}(0, T, N+1)$ 
7:    $A \leftarrow \text{zeros}(m-1, m-1)$ 
8:    $w_0 \leftarrow f(x[1:-1])$ 
9:    $w_1 \leftarrow (1 - \lambda^2)f(x[1:-1]) + \frac{\lambda^2}{2} (f(x[2:]) + f(x[0:-2])) + kg(x[1:-1])$ 
10:  for  $i \leftarrow 0$  to  $m-2$  do
11:     $A[i, i] \leftarrow 2(1 - \lambda^2)$ 
12:    if  $i < m-2$  then
13:       $A[i, i+1] \leftarrow \lambda^2$ 
14:       $A[i+1, i] \leftarrow \lambda^2$ 
15:    end if
16:  end for
17:  for each  $t_j$  in  $t[2:]$  do
18:     $w_j \leftarrow Aw_1 - w_0$ 
19:     $w_0 \leftarrow w_1$ 
20:     $w_1 \leftarrow w_j$ 
21:  end for
22:  return  $A, w_j, w_0, x$ 
23: end function
```

The corresponding implementation is available in the repository at <https://github.com/Julio-Medina/Finite-Difference-Method-Hyperbolic>.

2.2 Example

Use the finite-difference algorithm in Algorithm ??, with $h = 0.1$ and $k = 0.05$, to approximate the solution of the wave equation

$$\frac{\partial^2 u}{\partial t^2}(x, t) - 4 \frac{\partial^2 u}{\partial x^2}(x, t) = 0, \quad 0 < x < 1, \quad t > 0, \quad (29)$$

subject to the boundary conditions

$$u(0, t) = u(1, t) = 0, \quad t > 0,$$

and the initial conditions

$$u(x, 0) = \sin(\pi x), \quad \frac{\partial u}{\partial t}(x, 0) = 0, \quad 0 \leq x \leq 1. \quad (30)$$

The analytical solution is

$$u(x, t) = \sin(\pi x) \cos(2\pi t). \quad (31)$$

For $m = 10$, $N = 20$, and $\alpha = 2$, the grid spacings are $h = 0.1$ and $k = 0.05$, and the Courant number is $\lambda = 1$. The numerical and analytical values at $t = 1$ are compared in Table ??.

Table 1: Numerical and analytical solutions at $t = 1$.

i	x_i	$w_{i,20}$	$u(x_i, 1)$	$ u(x_i, 1) - w_{i,20} $
0	0.0	0.000000	0.000000e+00	0.000000e+00
1	0.1	0.309017	3.090170e-01	5.551115e-17
2	0.2	0.587785	5.877853e-01	2.220446e-16
3	0.3	0.809017	8.090170e-01	0.000000e+00
4	0.4	0.951057	9.510565e-01	3.330669e-16
5	0.5	1.000000	1.000000e+00	0.000000e+00
6	0.6	0.951057	9.510565e-01	2.220446e-16
7	0.7	0.809017	8.090170e-01	0.000000e+00
8	0.8	0.587785	5.877853e-01	0.000000e+00
9	0.9	0.309017	3.090170e-01	5.551115e-17
10	1.0	0.000000	1.224647e-16	1.224647e-16

3 Conclusions

The finite-difference method for hyperbolic partial differential equations with boundary conditions and initial displacement and velocity data has a relatively straightforward implementation. The method is based on constructing a tridiagonal matrix recurrence and advancing the numerical solution in time. The improved initialization formula preserves the second-order temporal accuracy of the centered scheme.

As shown in Table ??, the numerical solution agrees with the analytical solution to approximately machine precision for the selected example. This exceptionally small error is a special property of the chosen parameters: $\lambda = 1$, and the initial condition is a discrete sine eigenmode whose numerical frequency matches the exact frequency. In general, the centered scheme is second-order accurate in both time and space, provided the solution is sufficiently smooth and the CFL stability condition is satisfied.

References

- [1] Richard L. Burden and J. Douglas Faires, *Numerical Analysis*, 9th ed., Brooks/Cole, Cengage Learning, ISBN 978-0-538-73351-9.
- [2] Julio Medina, *Finite Difference Method for Elliptic Equations*, https://github.com/Julio-Medina/Finite_Difference_Method.
- [3] Richard S. Varga, *Matrix Iterative Analysis*, 2nd ed., Springer, DOI: 10.1007/978-3-642-05156-2.